

PRINCE CHAUHAN

SENIOR QUANTITATIVE ANALYST | SYSTEMATIC DERIVATIVES & STATISTICAL ARBITRAGE

New Delhi, India | +91-9650765343 | pchauhanrajput.pc@gmail.com | github.com/pchauhanrajputpc-sketch/quant-research-portfolio

SEBI Registered Research Analyst

PROFESSIONAL SUMMARY

Senior Quantitative Analyst and Systematic Derivatives Strategist with **9+ years of experience** designing, validating, and trading systematic options volatility and statistical arbitrage strategies across Indian index and single-stock derivatives. Specialized in implied volatility surface modeling (SVI, cubic splines), Johansen multivariate cointegration, unsupervised market regime detection, and microstructure-aware execution engines with strict risk and capital constraints. Combines production-grade Python research pipelines (Polars, NumPy, SciPy, Asyncio) with advanced financial accounting forensics and regulatory qualifications (CA Final Group 1, B.Com Hons 1st Div, SEBI Registered Research Analyst).

PROFESSIONAL EXPERIENCE

SMC Global Securities Ltd. - Senior Quantitative Analyst, Systematic Strategies & Quant Desk

Dec 2024 - Present

- Architected causal, event-driven simulation pipelines and point-in-time alpha research infrastructure on high-frequency tick and bar data for Nifty, Bank Nifty, Sensex, and single-stock derivatives, eliminating lookahead bias in intraday studies.
- Developed Bank Nifty basket statistical arbitrage models applying Johansen cointegration rank tests, dynamic rolling beta hedging, and Ornstein-Uhlenbeck spread half-life modeling across heavyweight banking equities to capture mean-reversion opportunities.
- Calibrated systematic options strategies delivering 2.5+ Out-of-Sample Sharpe and 2.1+ Profit Factor in strictly chronological simulations after pre-deducting conservative 0.50% round-trip execution friction (bid-ask spread crossing, market impact, statutory costs).
- Transitioned legacy rules-based execution to an asynchronous event-driven tick pipeline (Python Asyncio, broker WebSockets, memory-mapped IPC), improving fill efficiency by 24% and capturing 1.8 bps in bid-ask crossing savings.
- Partnered with proprietary risk desks to implement real-time portfolio Greeks monitoring (Delta, Gamma, Vega, Theta), dealer Gamma Exposure (GEX) mapping, parametric/historical VaR, automated volatility kill switches, and SPAN cross-margin capital limits.

Smart Bulls Capital - Research Analyst, Derivatives & Quantitative Research

Mar 2017 - Nov 2024

- Researched, structured, and traded systematic multi-leg options volatility strategies (calendar spreads, ratio butterflies, iron condors) with dynamic delta and vega rebalancing using real-time IV surface spline smoothing.
- Built Black-Scholes-Merton PDE inversion models (Newton-Raphson, Brent solvers), cubic spline & SVI strike smoothing, and real-time dealer Gamma Exposure (GEX) profiles to map institutional pin risk.
- Conducted fundamental equity research including financial statement forensics, discounted cash flow (DCF) modeling, and valuation across banking, energy, and industrial sectors to inform desk capital deployment to quant strategies.
- Managed daily P&L attribution, margin utilization, order slice execution, and drawdown limits across multi-strike options books, ensuring strict adherence to risk budgets and desk constraints.

Earlier Corporate Experience - Corporate Finance & Audit Roles

2014 - 2017

- Indian Oil Corporation Ltd. (IOCL) - Corporate Finance Executive (2016 - 2017): Managed treasury operations, working capital forecasting, and forensic variance audits.
- CA Matta & Associates - Finance & Audit Executive (2014 - 2016): Performed statutory audits, financial forensics, and corporate tax structuring.

QUANTITATIVE RESEARCH & ARCHITECTURE

- Point-in-Time Alpha Simulation Framework & Audit Gate:** Designed modular Python framework utilizing 7+ years of causal 1-minute historical data with zero lookahead bias; pre-deducts 0.50% turnover friction, enforces clearing-house SPAN margin models, portfolio leverage caps, 11-stage causal audit gate, 70/20/10 Walk-Forward Optimization, and 1,000-path Monte Carlo drawdown cones.
- Arbitrage-Free Options Volatility Surface & GEX Profiling:** Formulated Black-Scholes inversion using Newton-Raphson/Brent solvers, cubic spline & SVI surface fitting, arbitrage-free total variance constraints ($dW/dK \geq 0$), and dealer cross-margin Exposure (GEX) sensitivity profiling to stabilize IV surfaces and GEX maps across strikes/tenors.
- Bank Nifty Basket Cointegration & Statistical Arbitrage:** Developed Johansen multivariate cointegration rank tests, dynamic rolling beta weighting, Ornstein-Uhlenbeck spread half-life estimation, automated strike re-centering thresholds, and contract-rollover rules while maintaining strict delta neutrality.
- Market Regime Volatility Gating & Microstructure Simulator:** Engineered unsupervised K-Means clustering on Parkinson realized volatility, IV/RV ratios, and return skewness, reducing simulated maximum drawdown by 28%; paired with an execution simulator modeling order book queue priority and square-root market impact.

TECHNICAL & QUANTITATIVE COMPETENCIES

- Quantitative Research:** Statistical Arbitrage, Johansen Cointegration, Dynamic Beta Hedging, SVI Volatility Surface Fitting, Implied vs Realized Volatility (IV/RV, Parkinson), Market Regime Detection (K-Means), Walk-Forward Optimization (WFO), Monte Carlo Simulation.
- Derivatives & Risk Governance:** Options Pricing (Black-Scholes), Portfolio Greeks (Delta, Gamma, Vega, Theta), Delta/Vega Neutrality, Dealer Gamma Exposure (GEX), Parametric & Historical VaR, SPAN Cross-Margin Optimization, Portfolio Leverage Constraints.
- Programming & High-Performance Data:** Python 3.11+, Polars, DuckDB, NumPy, SciPy, Statsmodels, Scikit-Learn, VectorBT, SQL, Apache Parquet, Asyncio, WebSockets, Memory-Mapped IPC (mmap), ReportLab PDF Engine, Git, Linux/Bash.
- Markets & Microstructure:** NSE/BSE Equity & Index Derivatives (Nifty, Bank Nifty, Sensex), Single-Stock Futures & Options, Index Volatility Derivatives (VIX), Limit Order Book Dynamics, Smart Limit Slicing, Fill Slippage Minimization.

EDUCATION & REGULATORY CREDENTIALS

- Bachelor of Commerce (Honours):** Kirori Mal College, University of Delhi (2011 - 2014) | 71% Marks (First Division)
- Chartered Accountancy (CA Final):** Institute of Chartered Accountants of India (ICAI, 2018) | 78% Marks (Cleared Group 1)
- Regulatory Licenses:** SEBI Registered Research Analyst | NISM Series-XV: Research Analyst